16/06/2022

CONTEXT-FREE LANGUAGES

$L = L(G)$ G: context-free grammar ⇔ regular exp

$L = \{a^n b^n : n \geq 0\}$ ambiguous ✓

- Ambiguous grammar: If there is a string which has two different derivation trees or two leftmost derivations

- Chomsky Normal Form

First remove nullable variables, unit productions, useless variables (optional)

Then, for every symbol a:

New variable: T_a

Add production: $T_a \to a$

each production has form $A \to BC$ or $A \to a$

In productions with length at least 2 replace a with T_a

Productions of form $A \to a$ don't need to change

Replace any productions $A \to C_1 C_2 \dots C_n$

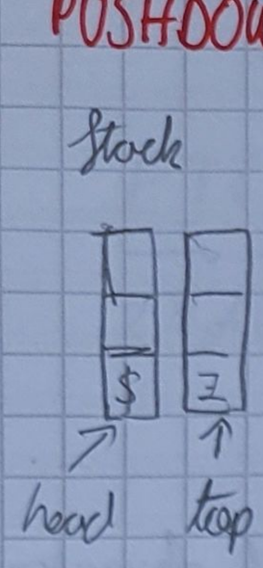
with $A \to C_1 V_1$

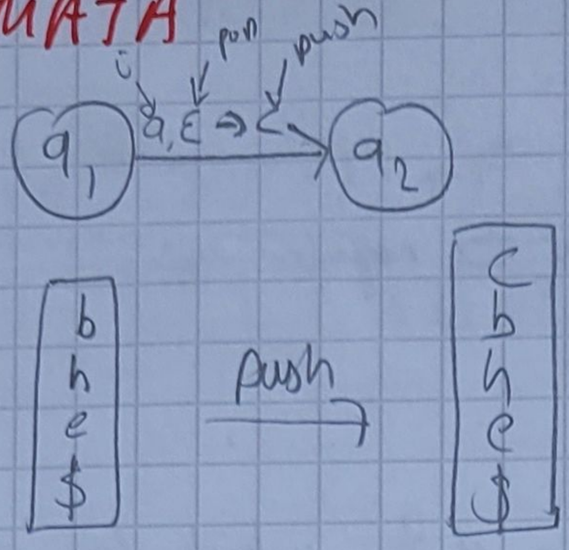
$V_1 \to C_2 V_2$

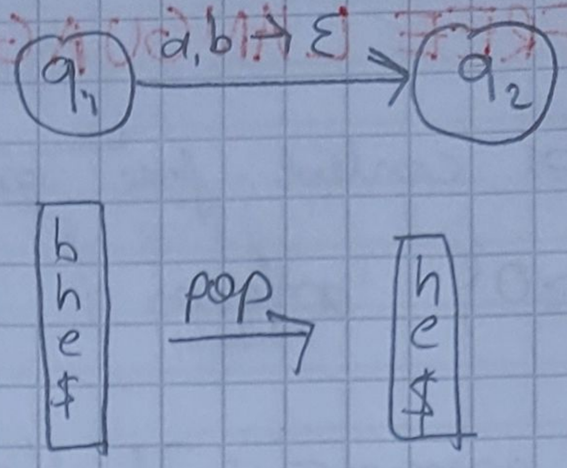
$\vdots$

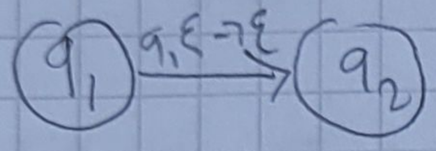
$V_{n-2} \to C_{n-1} C_n$

PUSHDOWN AUTOMATA





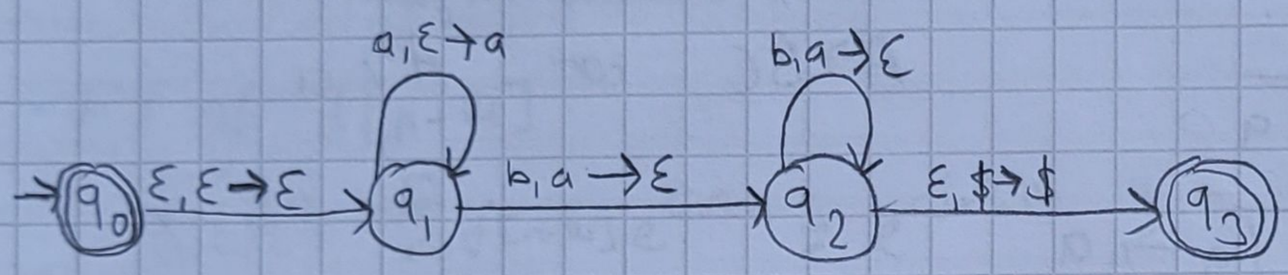




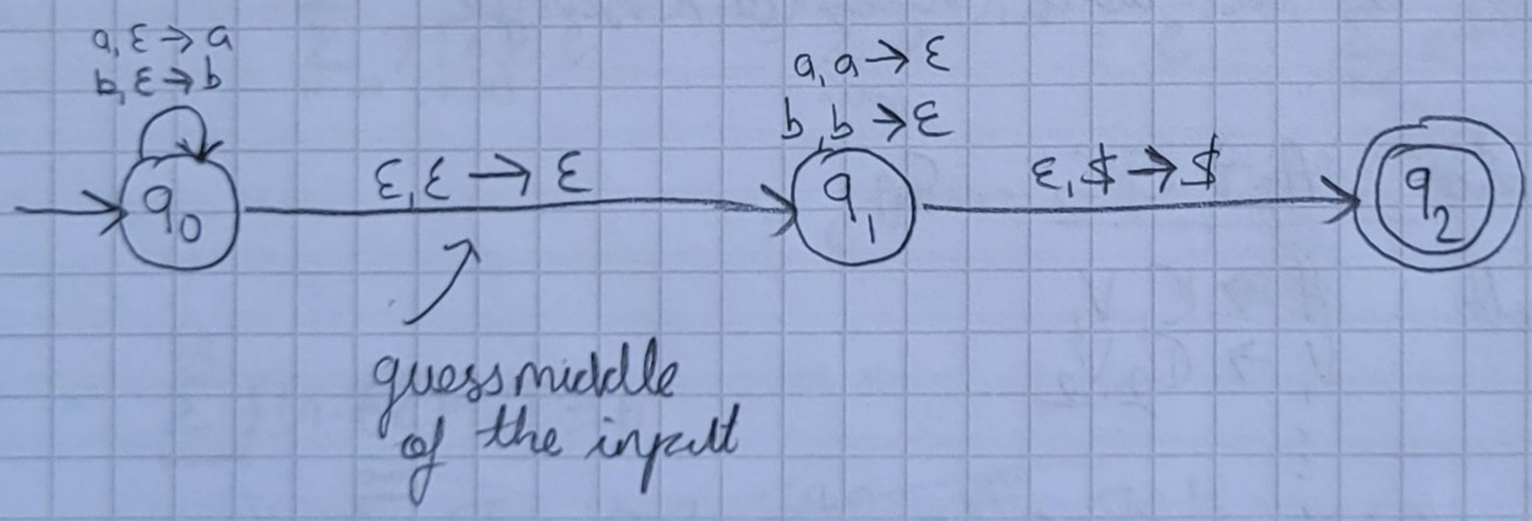
No change

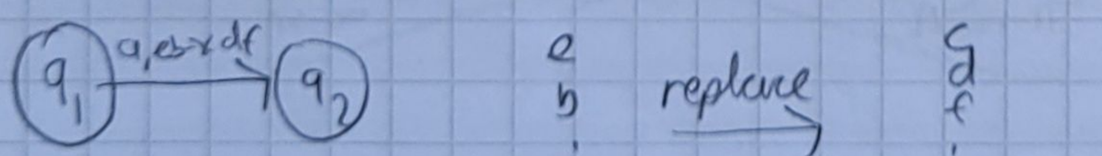
PDAs are non-deterministic

- $L(M) = \{a^n b^n : n \geq 0\}$ $S \to aSb$ $S \to \varepsilon$

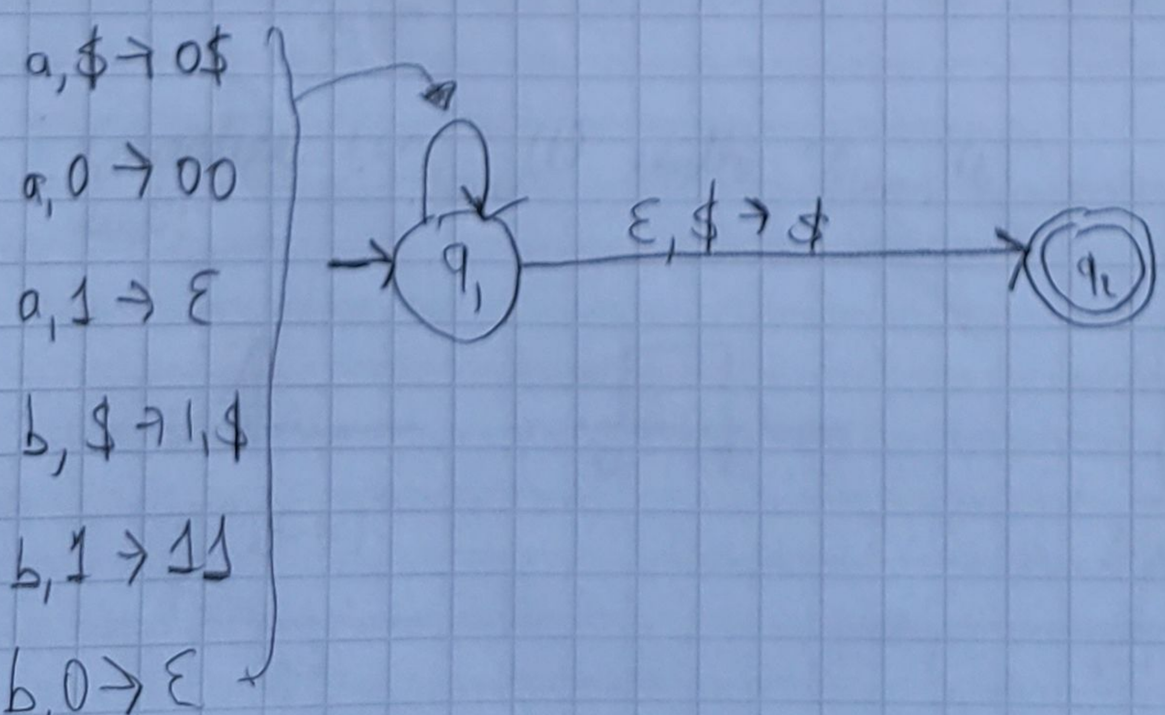


- $L(M) = \{vv^R : v \in \{a,b\}^*\}$ (palindrome)





- $L(M) = \{w \in \{a,b\}^* : n_a(w) = n_b(w)\}$



$M = (Q, \Sigma, \Gamma, \delta, q_0, z, F)$

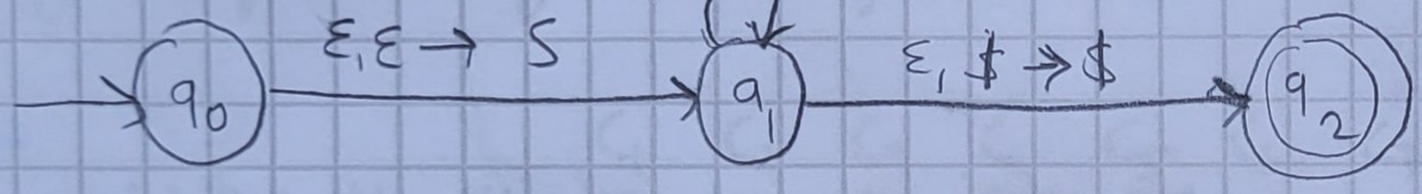
states — input alphabet — stack alphabet — transition function — initial state — stack start symbol — accept states

instantaneous description (q, u, s) q: current state u: remaining input s: current stack contents

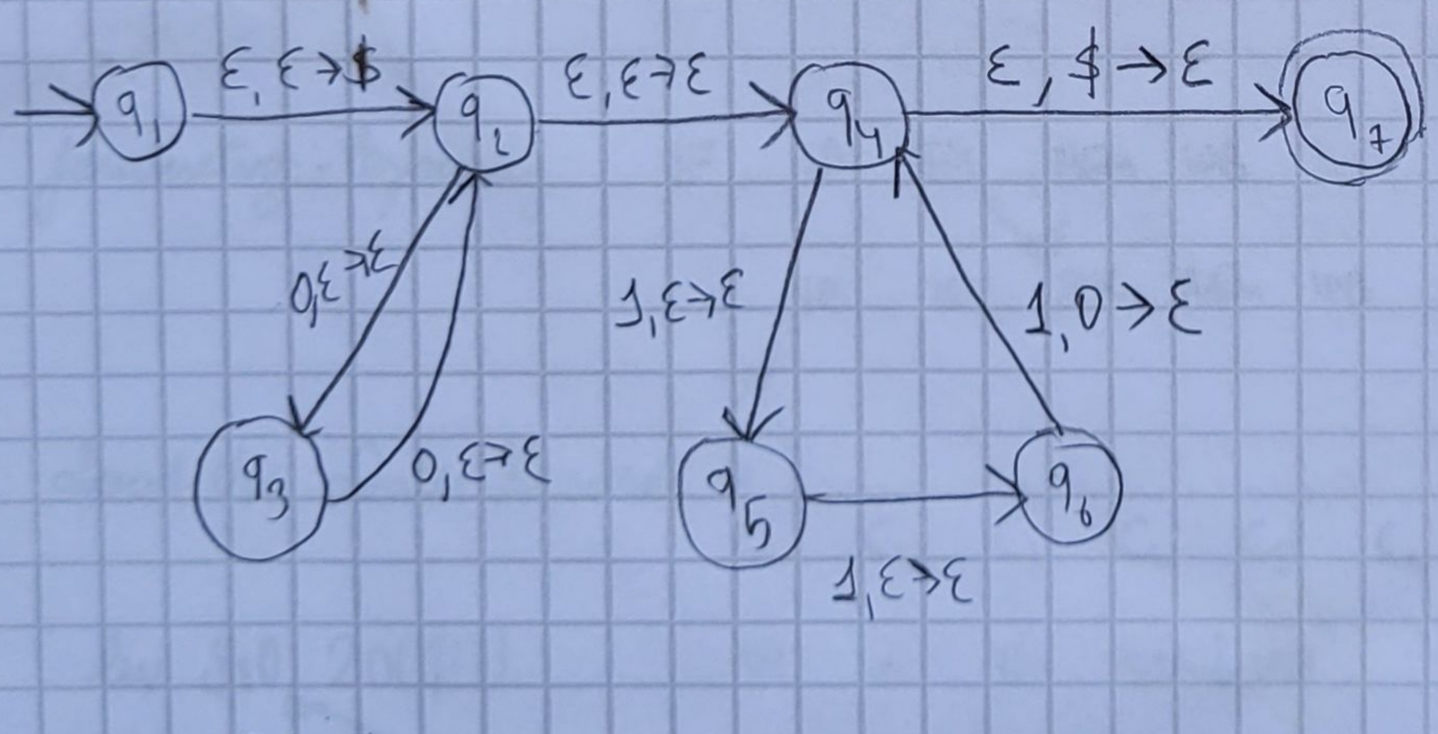
– PDA's recognizes context-free languages

Grammar: $S \to aSTb \quad S \to b \quad T \to Ta \quad T \to \varepsilon$ — wgcn !

$\varepsilon, S \to aSTb \quad \varepsilon, T \to \varepsilon$
$\varepsilon, S \to b \quad a, a \to \varepsilon$
$\varepsilon, T \to Ta \quad b, b \to \varepsilon$



- $L_1 = \{ 0^{2k} 1^{3k}, k \geq 0 \} \to \varepsilon, 00111, 000011111, \ldots$



- $L_2 = \{ 0^a 1^b 2^c \mid a, b, c \geq 0 \text{ and } a+b=c \}$

